INCH-POUND

MIL-PRF-19500/679
24 July 2000

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, SCHOTTKY
POWER RECTIFIER, SURFACE MOUNTED,
TYPE 1N6844U3, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, Schottky power rectifier in a surface mount package. Four levels of product assurance are provided as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (U3).

1.3 Maximum ratings.

Type	V_{RWM}	I_O (1) T_C = +125°C	I_{FSM} T_C =+25 °C T_p = 8.3 ms	C_J at 5 V dc	T_{STG}	T_J
	V	A dc	A (pk)	pF	°C	°C
1N6844U3	100	15	250	600	-65 to +150	-65 to +150

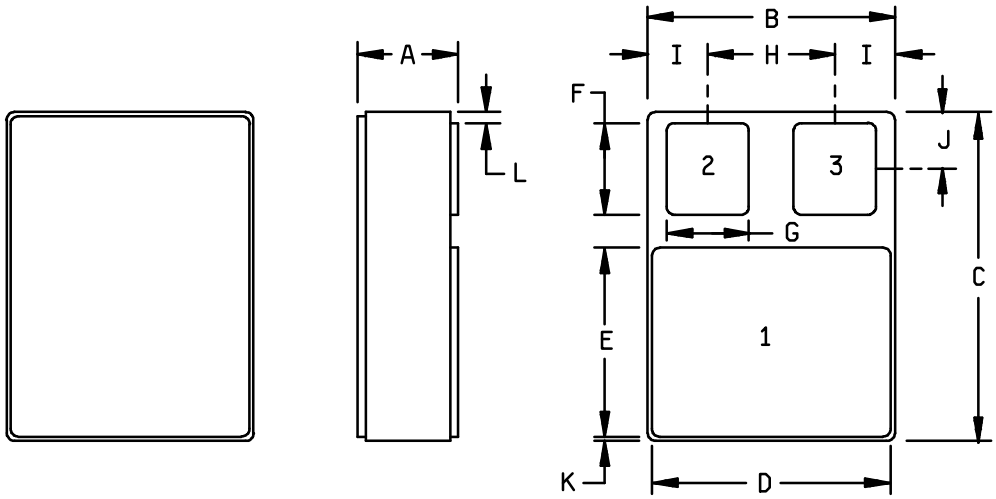
(1) Derate linearly at 600 mA/°C from $T_J = T_C$ = +125°C to +150°C.

1.4 Primary electrical characteristics. $R_{\theta JC}$ = 2.0°C/W maximum.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC/VAC, Post Office Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	0.111	0.122	2.82	3.10
B	0.291	0.301	7.39	7.65
C	0.395	0.405	10.03	10.29
D	0.281	0.291	7.14	7.39
E	0.220	0.230	5.59	5.84
F	0.115	0.125	2.92	3.18
G	0.090	0.100	2.29	2.54
H	0.125	0.135	3.68	3.94
I	0.073 TYP.		1.85 TYP.	
J	0.083 TYP.		2.11 TYP.	
K	0.005 TYP.		0.13 TYP.	
L	0.015 TYP.		0.015 TYP.	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Terminal 1 is cathode.
4. Terminals 2 and 3 are the anode.

SCHEMATIC

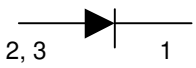


FIGURE 1. Physical dimensions and configuration (SMD.5, U3)

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and figure 1 herein.

3.4.1 Polarity. Polarity and terminal configuration shall be in accordance with figure 1 herein.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

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3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with table IX of MIL-PRF-19500, and table II herein. End point electrical measurements shall be in accordance with the applicable tests of table I, group A, subgroup 2 herein. Delta measurements shall be in accordance with table III herein.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 table IV and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c (1)	Method 3101 (see 4.3.2),	Method 3101 (see 4.3.2)
9 and 10	Not applicable	Not applicable
11	V_{F2} and I_{R1}	V_{F2} and I_{R1}
12	Condition A. See 4.3.1, t = 240 hours	Condition A. See 4.3.1, t = 48 hours
13	Subgroup 2 and 3 of table I herein; $\Delta V_{F2} = \pm 50$ mV, ΔI_{R1} = 100 percent of initial value or 25 µA, whichever is greater.	Subgroup 2 of table I herein; $\Delta V_{F2} = \pm 50$ mV, ΔI_{R1} = 100 percent of initial value or 25 µA, whichever is greater.

(1) Shall be performed anytime after screen 3.

4.3.1 Power burn-in conditions. Burn-in conditions are as follows:

MIL-STD-750 method 1038, test condition A. T_C = +125°C to +150°C; V_R = 80 percent of rated V_{RWM} .

4.3.2 Thermal impedance ($Z_{\theta JX}$ measurements) The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit and conditions for $Z_{\theta JX}$ in screening (table IV of MIL-PRF-19500) shall be derived by each vendor by means of process control of actual measurements which characterizes the die attach process. When three lot date codes have exhibited control, the data from these three lots will be used to establish a fixed screening limit (not exceed the group A limit). Once a fixed limit has been established, monitor all future sealing lots using a sample from each lot to be plotted on the applicable SPC chart.

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4.3.2.1 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101 (read and record data $Z_{\theta JX}$) derived conditions, limits, and thermal response curve shall be supplied to the qualifying activity on the qualification lot prior to qualification approval.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table V of MIL-PRF-19500, and table I herein. Thermal impedance $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables VIa (JANS) and VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and paragraphs 4.4.2.1 and 4.4.2.2 herein. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein. Delta measurements shall be in accordance with table III herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	4066	Condition A, I_{FSM} = 250 A superimposed on I_O = 10 A, 1 surge of 8.3 ms.
B4	1037	I_F or I_O = 2 A (minimum) ΔT_J = +85 °C, + 15°C, - 5 °C for 2,000 cycles minimum.
B5	1038	Condition B, T_J = +150°C, T_A = +100°C, 240 hrs.
B6	3101	Limit for thermal resistance is 2.0°C/W.

4.4.2.2 Group B inspection, table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B2	4066	Condition A, I_{FSM} = 250 A superimposed on I_O = 10 A; 1 surge of 8.3 ms.
B3	1037	I_F or I_O = 2 A (minimum); ΔT_J = +85°C, +15°C, -5°C for 2,000 cycles minimum.
B5	----	Not applicable.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein. Delta measurements shall be in accordance with table III herein.

Subgroup	Method	Condition
C2	2036	Terminal strength is not applicable.
C5	3101	$R_{\theta JC}$ = 2.0 °C/W maximum.
C6	1037	I_F or I_O = 2 A (minimum); $T_J \leq +150°C$, $T_A = +100°C$ for 6,000 cycles minimum.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance shall be measured as follow in accordance with MIL-STD-750, method 3101.

a. I_M.....10 mA

b. I_H.....10 - 15 A

c. t_{MD}.....100 µs maximum

d. t_H heating time.....10 ms

e. $R_{\theta JC}$.....2.0°C/W

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TABLE I. Group A inspection. 1/

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance	3101	See 4.4.1	$Z_{\theta JX}$		1.9	°C/W
Forward voltage	4011	I_{FM} = 5 A (dc) pulsed (see 4.5.1)	V_{F1}		0.70	V dc
Forward voltage	4011	I_{FM} = 15 A (dc) pulsed (see 4.5.1)	V_{F2}		0.90	V dc
Forward voltage	4011	I_{FM} = 20 A (dc) pulsed (see 4.5.1)	V_{F3}		1.0	V dc
Reverse current leakage 1N6844	4016	DC method, (see 4.5.1.) V_R =100 V dc	I_{R1}		50	µA dc
Subgroup 3						
High temperature operation:		T_A = +125°C				
Reverse current leakage 1N6844	4016	DC method, pulsed (see 4.5.1) V_R =100 V dc	I_{R2}		10	mA dc
Forward voltage	4011	I_F = 5 A pulsed (see 4.5.1)	V_{F4}		0.58	V dc
Forward voltage	4011	I_F = 15 A pulsed (see 4.5.1)	V_{F5}		0.72	V dc
Low temperature operation:		T_A = -55°C				
Forward voltage	4011	Pulsed (see 4.5.1) I_F = 5 A (dc)	V_{F6}		0.85	V dc
Subgroup 4						
Junction capacitance	4001	V_R = 5 V dc; f = 1 MHz, V_{SIG} = 50 mV (p-p) (max)	C_J		600	pF
Subgroup 5						
Dielectric withstand	1016	V_R = 600 V dc; From lid to bottom case. All terminals shorted.	I_{R3}		10	µA dc
Subgroup 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification Inspection
	Method	Conditions	
Subgroup 1			38 devices, c = 0
Thermal shock (temperature cycling)	1051	500 cycles. Condition G	
Hermetic seal	1071		
Electrical measurements		See table I, group A, subgroup 2 and steps 1 and 2 of table III herein	
Subgroup 2			38 devices, c = 0
Steady-state blocking life	1038	t = 1,000 hours, T_J = + 125 °C; V_R = 80% of rated V_{RWM}	
Electrical measurements		See table I, group A, subgroup 2 and steps 1 and 2 of table III herein	
Subgroup 3			
Not applicable			
Subgroup 4			5 devices, c = 0
Thermal resistance	3101	See 4.5.2, $R_{\theta JC}$ = 2. 0 °C/W	
Subgroup 5			5 devices, c = 0
Surge	4066	Condition A, T_A = +25°C I_{FSM} = 250 A, 1 surge of 8.3 ms. V_R = 0;	
Electrical measurements		See table I, group A, subgroup 2 and steps 1, 2, and 3 of table III herein	

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TABLEIII. Groups B, C, and E, delta measurement 1/ 2/ 3/ 4/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Forward voltage	4011	I_F = 10 A (pk) pulsed (see 4.5.1)	ΔV_{F2}		±50	mV
2	Reverse current leakage	4016	pulsed (see 4.5.1) DC method $V_R = V_{RWM}$	ΔI_{R1}		25 µA or 100 % of initial reading, which ever is greater.	
3	Thermal impedance	3101	See 4.4.1	$Z_{\theta JX}$		1.9	°C/W

1/ The delta measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroup 3, see table III herein, steps 1 and 2.

b. Subgroup 4, see table III herein, steps 1, 2, and 3.

c. Subgroup5, see table III herein, steps 1, 2, and 3.

2/ The delta measurements for table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1 and 2.

b. Subgroup 3, see table III herein, steps 1, 2, and 3.

c. Subgroup 6, see table III herein, steps 1 and 2.

3/ The delta measurements for table VII of MIL-PRF-19500 are as follows:

a. Subgroup 2 and 3, see table III herein, steps 1 and 2 for all levels.

b. Subgroup 6, see table III herein, steps 1, 2 and 3 for all levels.

4/ The delta measurements for table IX of MIL-PRF-19500 are as follows:

a. Subgroups 1 and 2, see table III herein, steps 1 and 2 for all levels.

b. Subgroup 5, see table III herein, steps 1, 2 and 3 for all levels.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1).

c. Type designation and product assurance level.

d. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers' List (QML) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43216-5000.

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/679	2. DOCUMENT DATE 24 July 2000

3. **DOCUMENT TITLE** SEMICONDUCTOR DEVICE, DIODE, SILICON, SCHOTTKY POWER RECTIFIER, SURFACE MOUNTED, TYPE 1N6844U3, JAN, JANTX, JANTXV, and JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial 614-692-0510	DSN 850-0510	FAX 614-692-6939	EMAIL Alan_Barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus, ATTN: DSCC-VAC P.O. Box 3990 Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533 Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888			

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